MODEL 31 SERIES

008360
Issue 2

Low, Medium, and High Range Precision Miniature Load Cell

DESCRIPTION

The Honeywell Model 31 series of precision miniature load cells are designed to measure both tension and compression forces across a wide range—from 500 g (5 N) to 10,000 lb. These rugged, high-accuracy load cells offer exceptional performance in compact packages and are built for applications where space is limited and accuracy is critical.

All Model 31 units feature a welded 17-4 PH stainless steel construction, engineered to minimize the effects of off-axis loading and ensure long-term stability—particularly in ranges 1,000 g and above. A modification is available for fully welded units suitable for underwater use. Each unit includes male threaded attachments and provides high accuracy from 0.15 % to 0.25 % full scale.

The low-range models measure forces up to 500 g (5 N) and are ideal for highly sensitive applications requiring precise, small-scale force measurements.

The mid-range models, spanning 1,000 g to 1,000 lb, incorporate an electrical balance module in the lead wire (approximately 1 in × 0.087 in thick) for enhanced signal conditioning. This module does not need to be at the same temperature as the transducer, adding flexibility in varied environments.

The high-range models are built for robust force measurements from 2,000 lb to 10,000 lb, maintaining the same compact footprint and accuracy expected from the Model 31 series.

FEATURES

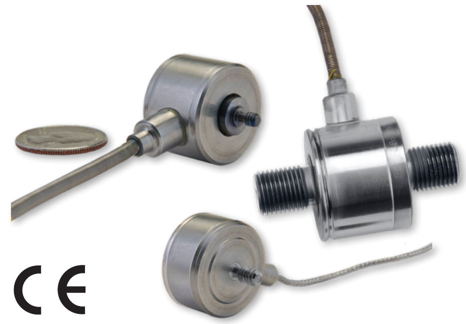
- Low range: 50 g/0.5 N to 500 g/5 N
- Mid range: 1000 g/10 N to 1000 lb/5 kN
- High range: 2000 lb to 10,000 lb/10 kN to 50 kN
- mV/V output
- Stainless steel
- Miniature design
- Low range: Double diaphragm construction
- High range: Stabilized column construction

MECHANICAL AND ENVIRONMENTAL SPECIFICATIONS

- Operating temperature : -53°C to 121°C [-65°F to 250°F]
- Low range: double diaphragm construction
- High range: stabilized column construction
- High-range maximum allowable load 150 %FS

PORTFOLIO

From general purpose load cells to fatigue-rated, high performance products, Honeywell offers a comprehensive selection of tension, compression, and universal measurement load cells to meet market demands. Each of our load cells can be customized to meet your needs, whatever your application. To view the entire product portfolio, click here.



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APPLICATIONS

- Cable tension
- Industrial process control
- Medical control systems
- Medical equipment testing
- Pharmaceutical process or product control
- Semiconductor/electronics testing
- Aerospace testing

DIFFERENTIATION

- Enhanced accuracy of 0.15 % to 0.2 %
- Newton, gram, and pound force ranges available
- Tension and compression special calibration

ORDERING INFORMATION

- Order code AL311
- Standard 2u. Unamplified, mV/V output. Compatible in-line amplifiers available upon request.
- Consult factory & Honeywell Sales for custom build requests

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MODEL 31 SERIES
PRECISION MINIATURE LOAD CELLS

TABLE 1. MODEL 31 LOW-RANGE PERFORMANCE SPECIFICATIONS

Characteristic	Measure
Load ranges	50 g, 150 g, 250 g, 500 g 0.5 N, 1.5 N, 2.5 N, 5 N
Linearity	±0.15 % full scale
Hysteresis	±0.15 % full scale
Non-repeatability	±0.1 % full scale
Full scale output (tolerance)	
50 g to 150 g	0.1 mV/V/g (max.)
0.5 N to 1.5 N	10 mV/V/N (max.)
250 g/2.5 N to 500 g/5 N	20 mV/V (nominal)
Operation	Tension/compression[3]
Resolution	Infinite

TABLE 2. MODEL 31 MID-RANGE PERFORMANCE SPECIFICATIONS

Characteristic	Measure
Load ranges	1000 g, 5 lb, 10 lb, 25 lb, 50 lb, 100 lb, 250 lb, 500 lb, 1000 lb 10 N, 20 N, 50 N, 100 N, 200 N, 500 N, 1 kN, 2 kN, 5 kN
Non-linearity	
1000 g to 250 lb 10 N to 1 kN	±0.15 % full scale
500 lb to 1000 lb 2 kN to 5 kN	±0.2 % full scale
Hysteresis	
1000 g to 250 lb 10 N to 1 kN	±0.15 % full scale
500 lb to 1000 lb 2 kN to 5 kN	±0.2 % full scale
Non-repeatability 1000 g	
1000 g, 10 N	±0.1 % full scale
5 lb to 1000 lb 20 N to 5 kN	±0.05 % full scale
Output (tolerance) 1000 g	
1000 g, 10 N	1.5 mV/V (nominal)
5 lb to 1000 lb 20 N to 5 kN	2 mV/V (nominal
Operation	Tension/compression[3]
Resolution	Infinite

TABLE 3. MODEL 31 HIGH-RANGE PERFORMANCE SPECIFICATIONS

Characteristic	Measure
Load ranges	2000 lb, 5000 lb, 10000 lb 10 kN, 20 kN, 50 kN
Linearity	±0.2 % full scale
Hysteresis	±0.2 % full scale
Non-repeatability	±0.05 % full scale
Full scale output (tolerance)	2 mV/V (nominal)
Operation	Tension/compression[3]
Resolution	Infinite

TABLE 4. ENVIRONMENTAL SPECIFICATIONS

Characteristic	Measure
Temperature, operating	-53°C to 121°C [-65°F to 250°F]
Temperature, compensated	15°C to 71°C [60°F to 160°F]
Temperature, storage	-73°C to 148°C [-100°F to 300°F]
Temperature effect, zero	±0.01 % full scale/°C [±0.005 % full scale/°F]
Temperature effect, zero (low range)	±0.03 % full scale/°C [±0.015 % full scale/°F]
Temperature effect, span	±0.01 % reading/°C [±0.005 % reading/°F]
Temperature effect, span (low range)	±0.03 % reading/°C [±0.015 % reading/°F]

TABLE 5. ELECTRICAL SPECIFICATIONS

Characteristic	Measure
Low-range strain gauge type	Semiconductor
Mid-range/high-range strain gauge type	Bonded foil
Low-range excitation (calibration)	5 Vdc
Mid-range excitation (calibration)	
1 kg to 10 lb 10 N to 50 N	5 Vdc
25 lb to 1000 lb 100 N to 5 kN	10 Vdc
High-range excitation (calibration)	10 Vdc
Insulation resistance	5000 Mohm @ 50 Vdc
Bridge resistance (mid/high range)	350 ohm
Bridge resistance (low range)	500 ohm (nominal)
Zero balance	±1 % full scale max.
Electrical termination	see Figures 2 to 5

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TABLE 6. MECHANICAL SPECIFICATIONS	
Characteristic	**Measure**
Low-range allowable load	20 N [5 lb][1]
Mid- & High-range maximum allowable load	150 %FS[1]
Material	17-4 PH stainless steel

TABLE 7. WIRING CODE	
Pin	**Designation**
Red	(+) excitation
Black	(-) excitation
Green	(-) output
White	(+) output

TABLE 8. DEFLECTIONS & RINGING FREQUENCIES			
Capacity	**Deflection at full scale**	**Ringing frequency**	**Weight**
50 g to 500 g 0.5 N to 5 N	0,02 mm [0.0008 in]	740 Hz	90 g
1000 g to 10 lb 10 N to 50 N	0,03 mm [0.001 in]	3000 Hz	21 g
25 lb to 100 lb 100 N to 500 N	0,03 mm [0.001 in]	10000 Hz	63 g
250 lb to 1000 lb 1 kN to 5 kN	0,04 mm [0.0015 in]	12000 Hz	80 g
2000 lb 10 kN	0,03 mm [0.001 in]	26000 Hz	60 g
5000 lb 20 kN	0.04 mm [0.0015 in]	21000 Hz	125 g
10000 lb 50 kN	0,04 mm [0.0015 in]	17000 Hz	250 g

Figure 1. Typical System Diagram

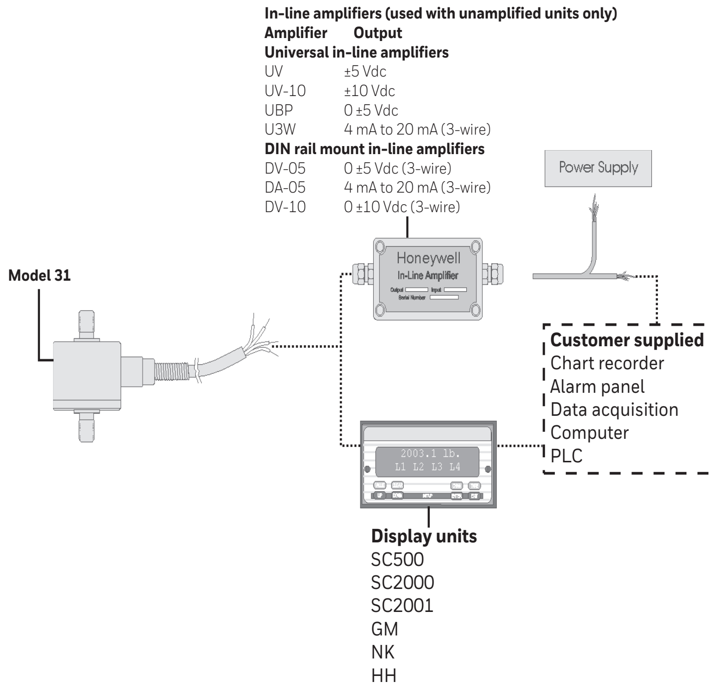

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TABLE 9. MOUNTING DIMENSIONS

Ranges	Thread (T)	D	H	C	F	A	B
50 g, 150 g, 250 g, 500 g 0.5 N, 1.5 N, 2.5 N, 5 N	#6 - 32 UNC M4 × 0.7	Ø25,4 mm [1.00 in]	21,8 mm [0.86 in]	6,4 mm [0.25 in]	2,8 mm [0.11 in]	12,7 mm [0.50 in]	9,7 mm [0.38 in]
1000 g, 5 lb, 10 lb 10 N, 20 N, 50N	#6-32 UNC M4 × 0.7	Ø19 mm [0.75 in]	12,5 mm [0.49 in]	6,4 mm [0.25 in]	1,2 mm [0.05 in]	7,87 mm [0.31 in]	4,83 mm [0.19 in]
25 lb, 50 lb, 100 lb 100 N, 200 N, 500 N	#10-32 UNF M5 × 0.8	Ø25,4 mm [1.00 in]	15,5 mm [0.61 in]	6,4 mm [0.25 in]	2,3 mm [0.09 in]	12,7 mm [0.50 in]	6,35 mm [0.25 in]
250 lb, 500 lb, 1000 lb 1 kN, 2 kN, 5 kN	1/4-28 UNF M6 × 1	Ø25,4 mm [1.00 in]	13,21 mm [0.52 in]	9,7 mm [0.38 in]	0,8 mm [0.03 in]	12,7 mm [0.50 in]	6,35 mm [0.25 in]
2000 lb 10 kN	3/8-24 UNF M10 x 1.5	Ø25.4 mm [1.00 in]	19,1 mm [0.75 in]	12,7 mm [0.50 in]	0,3 mm [0.01 in]	12,7 mm [0.50 in]	9,7 mm [0.38 in]
5000 lb 20 kN	1/2-20 UNF M12 x 1.5	Ø 31,8 mm [1.25 in]	25.4 mm [1.00 in]	16 mm [0.63 in]	0,3 mm [0.01 in]	12,7 mm [0.50 in]	9,7 mm [0.38 in]
10000 lb 50 kN	3/4-16 UNF M20 x 1.5	Ø 35,05 mm [1.38 in]	28,7 mm [1.13 in]	22,35 mm [0.88 in]	0,3 mm [0.01 in]	12,7 mm [0.50 in]	9,7 mm [0.38 in]

Figure 2. Model 31 Mounting Dimension
50 gm to 500 gm / 0.5 N to 5N

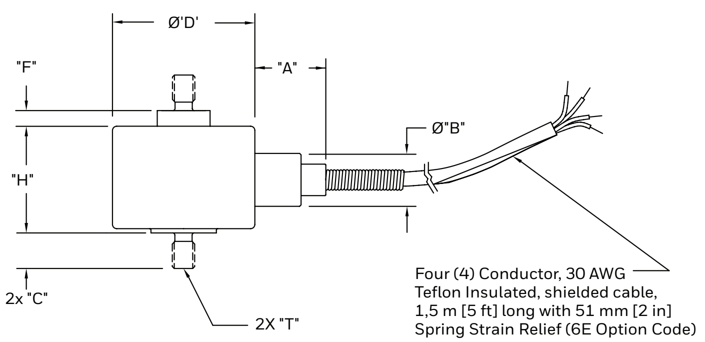

Figure 3. Model 31 Mounting Dimension
10 N to 50 N / 1 kg to 10 lb (6am termination required)

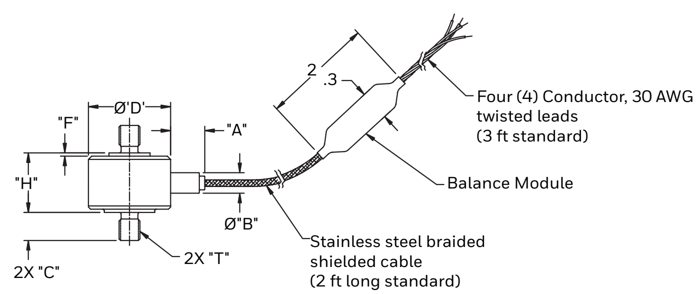

Figure 4. Model 31 Mounting Dimension
100 N to 5K kN / 25 lb to 1000 lb (6E termination required)

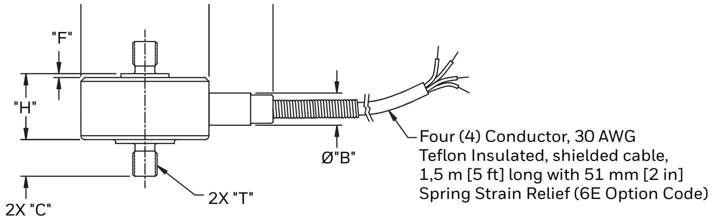

Figure 5. Model 31 Mounting Dimension
10 kN to 50 kN / 2000b to 10000 lb
(6E termination required)

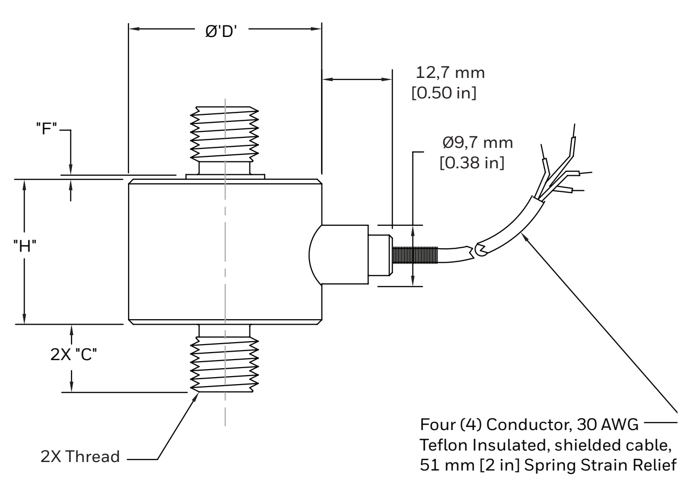

Figure 6. Product Nomenclature

AL311 MK, 1A, 2U, 6E, 9A, 15C, 32B

LOAD TYPE
AL311 – Model 31 Load Cell

RANGE

AJ – 50 g	MD – 0.5 Newton
AL – 150 g	NJ – 1.5 Newton
AN – 250 g	NK – 2.5 Newton
AP – 500 g	MG – 5 Newton
AR – 1,000 g	MH – 10 Newton
AT – 5 lb	MI – 20 Newton
AV – 10 lb	MK – 50 Newton
BL – 25 lb	ML – 100 Newton
BN – 50 lb	MM – 200 Newton
BR – 100 lb	MY – 500 Newton
CN – 250 lb	MN – 1,000 Newton
CR – 500 lb	MO – 2,000 Newton
CV – 1,000 lb	MQ – 5,000 Newton
DL – 2,000 lb	MR – 10,000 Newton
DR – 5,000 lb	MS – 20,000 Newton
DV – 10,000 lb	MT – 50,000 Newton

TEMPERATURE COMPENSATION
1A – 15°C to 71°C [60°F to 160°F]
1C – -18°C to 85°C [0°F to 185°F]
1H – 21°C to 204°C [70°F to 400°F][8]

INTERNAL AMPLIFIER
2U – Unamplified, mV/V output

OVERLOAD STOPS
4A – Overload stop[7]

ELECTRICAL TERMINATION
6AM – Teflon® integral cable] with balance module[7]
6E – Teflon® integral cable
6I – Integral underwater cable, 3 m [10 ft][6]

ADDITIONAL POINT CALIBRATION
– Standard 5-point calibration None
9A – 10 point calibration[9]

ELECTRICAL CONN. ORIENTATION
15C – Radial exit

SPECIAL CALIBRATION[3]
– Tension (+) only (None)
30B – Tension (+) and compression (-)
30C – Tension (-) and compression (+)

MOUNTING THREADS
– #6-32 UNC (none)
32B – M4 × 0.7
32C – M5 × 0.8
32D – M6 × 1
32F – M10 × 1.5
32G – M12 × 1.5
32I – M20 × 1.5

CALIBRATION MEMORY[4]
– No storage (None)
53S – Phoenix connector with Signature Calibration Module
53T – TEDS IEEE 1451.4 module

Mandatory for Load Cell Order Code
Optional for Load Cell Order Code

NOTES

1. Allowable maximum loads - maximum load to be applied without damage.[2]
2. Without damage - loading to this level will not cause excessive zero shift or performance degradation. The user must consider fatigue life for long term use and structural integrity. All structurally critical applications (overhead loading, etc.) should always be designed with safety redundant load paths.
3. Standard calibration for tension/ compression load cells is in tension only.
4. TEDS IEEE 1451.4 module installed at end of cable.
5. Maximum operating temperature for options 53s and 53t is 85°C [185°F]
6. Option 6i only available on mid-range/ high-range and may increase the load cell height and/or diameter. Consult factory
7. Option 4a and 6am is only availble with option AR, AT, AV, MH, MI, MK and must be included in order code string
8. Option 1h only available with mid-range-high-range only

9 Option 9A not available with Low-range

For more information

Honeywell Sensing Solutions services its customers through a worldwide network of sales offices and distributors. For application assistance, current specifications, pricing, or the nearest Authorized Distributor, visit automation.honeywell.com or call:

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